Chinese Remainder Theorem

An End-to-End Meta-Math Walkthrough

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Abstract

Build a selector for each congruence, combine the selectors, and prove that two solutions differ by the product of the pairwise coprime moduli. This report presents a complete proof relative to explicit imports, a dependency graph, versioned node contracts, five-component activities, exact examples, and an illustrative repair. The interactions are authored explanations, not a transcript of deployed agents or a proof-assistant certificate.

Project NT-03 Report version 1 Number Theory

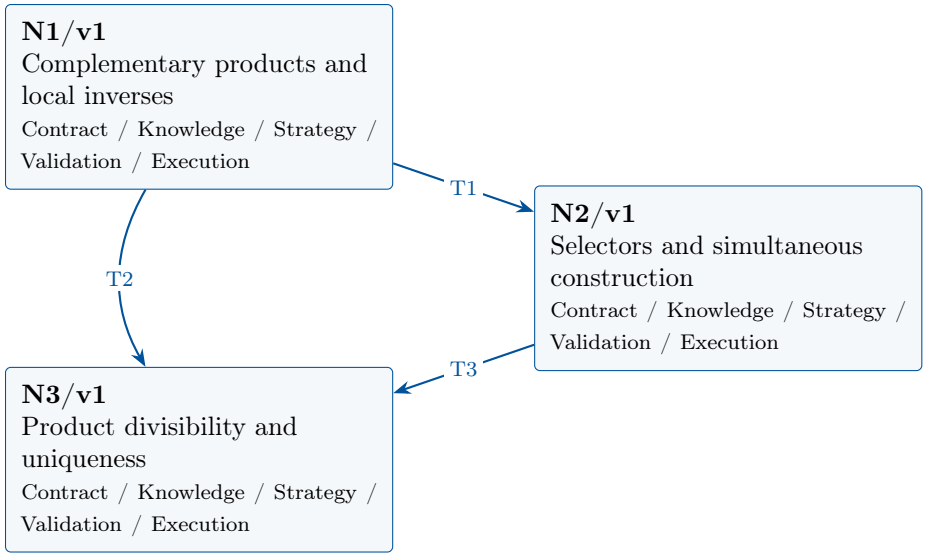
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let $k \geq 1$, let $m_1, \ldots, m_k \geq 2$ be pairwise coprime integers, and let $r_1, \ldots, r_k \in \mathbb{Z}$. With $M = \prod_i m_i$, the system $x \equiv r_i \pmod{m_i}$ has exactly one solution class modulo M, and an explicit integer representative can be constructed.*

A. Foundations and permitted imports

1. NT-01/N5, v1: if $\gcd(u, m) = 1$ with $m \geq 2$, Bézout constructs $t, s \in \mathbb{Z}$ such that $tu + sm = 1$.
2. NT-02/N1 and N2, v1: every integer greater than 1 has a prime divisor, and a prime dividing a finite product divides one factor. These justify coprimality of complementary and partial products. Empty products equal 1.

II. NODE INTERACTION WORKFLOW



Blue arrows are accepted mathematical dependencies, with exact transfers T1–T3 below. Repair messages are described separately; they are not premises of the proof.

A. Versioned evidence-transfer ledger

T1: N1/E1 v1 to N2.2–3
Claim. For $M_i = M/m_i$, $\gcd(M_i, m_i) = 1$ and integers t_i, s_i satisfy $t_i M_i + s_i m_i = 1$. Any product of distinct other moduli is also coprime to m_i.
Compatibility. All selectors use their matching modulus and inverse certificate.
Evidence. Node N1 activities below and the continuous proof in Section IV.

T2: N1/E1 v1 to N3.2
Claim. For $M_i = M/m_i$, $\gcd(M_i, m_i) = 1$ and integers t_i, s_i satisfy $t_i M_i + s_i m_i = 1$. Any product of distinct other moduli is also coprime to m_i.
Compatibility. The transferred export includes partial-product coprimality, not just inverse values.
Evidence. Node N1 activities below and the continuous proof in Section IV.

T3: N2/E2 v1 to N3.3
Claim. $x = \sum_i r_i M_i t_i$ satisfies $x \equiv r_j \pmod{m_j}$ for every j.
Compatibility. At least one simultaneous solution exists for the fixed residue tuple.
Evidence. Node N2 activities below and the continuous proof in Section IV.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Complementary products and local inverses

Object and role. Construct one certified unit for each modulus and retain the product-coprimality lemma needed by uniqueness.

Inputs. Pairwise coprime $m_i \geq 2$; NT-01 and the declared NT-02 lemmas.

C — Contract

Publish each local inverse with its own modulus; retain the coprimality lemma in the same export.

K — Knowledge

Retain the indexed moduli and prime-divisor/finite-product lemmas.

S — Strategy

A shared prime with a complementary product would divide a different modulus, contradicting pairwise coprimality.

V — Validation

The gcd of all moduli being 1 is weaker than pairwise coprimality. For $k = 1$, $M_1 = 1$.

E — Execution

Set $M = \prod_i m_i$ and $M_i = M/m_i$. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N1.1 Set $M = \prod_i m_i$ and $M_i = M/m_i$.

N1.2 If a common divisor with m_i exceeded 1, choose a prime divisor and use Euclid's lemma to contradict pairwise coprimality; the same argument applies to any partial product.

N1.3 Use Bézout to find $t_i M_i + s_i m_i = 1$ for every i.

Export E1/v1. For $M_i = M/m_i$, $\gcd(M_i, m_i) = 1$ and integers t_i, s_i satisfy $t_i M_i + s_i m_i = 1$. Any product of distinct other moduli is also coprime to m_i.

Acceptance gate. Every local inverse and every partial-product coprimality claim refers to the original indexed moduli.

Targeted repair. If pairwise coprimality fails, stop this construction. A generalized CRT would need a separately stated contract; recheck N2 and N3.

B. N2/v1: Selectors and simultaneous construction

Object and role. Combine independent local certificates into one integer satisfying every congruence.

Inputs. N1/E1 and arbitrary integer residues r_i.

C — Contract

Publish existence only after checking each coordinate.

K — Knowledge

Retain inverse certificates with their indices and the divisibility $m_j \mid M_i$ for $i \neq j$.

S — Strategy

Each selector $M_i t_i$ is 1 at its own modulus and 0 at every other modulus.

V — Validation

Residues need not be units; distinguish the surviving term from the vanishing cross-terms.

E — Execution

Define $e_i = M_i t_i$ and $x = \sum_i r_i e_i$. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N2.1 Define $e_i = M_i t_i$ and $x = \sum_i r_i e_i$.

N2.2 Fix j. Terms with $i \neq j$ vanish modulo m_j.

N2.3 The remaining term is r_j because $M_j t_j \equiv 1 \pmod{m_j}$.

Export E2/v1. $x = \sum_i r_i M_i t_i$ satisfies $x \equiv r_j \pmod{m_j}$ for every j.

Acceptance gate. The coordinate check is valid for an arbitrary index j, hence for all congruences.

Targeted repair. An incorrect selector returns to its N1 inverse certificate; rebuild the sum and recheck N3.

C. N3/v1: Product divisibility and uniqueness

Object and role. Prove the exact modulus of uniqueness and identify the full solution set.

Inputs. N1/E1 partial-product coprimality; N2/E2; two solutions x, y.

C — Contract

Publish uniqueness of a residue class, without claiming a unique integer.

K — Knowledge

Retain $m_i \mid (x - y)$ and Bézout for coprime factors.

S — Strategy

Combine divisibility one coprime factor at a time.

V — Validation

Individual divisibility by noncoprime integers does not imply divisibility by their product.

E — Execution

If coprime a, b divide d, write $d = ak$ and $ua + vb = 1$. Multiplying by k shows $b \mid k$, hence $ab \mid d$. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N3.1 If coprime a, b divide d, write $d = ak$ and $ua + vb = 1$. Multiplying by k shows $b \mid k$, hence $ab \mid d$.

N3.2 Apply this repeatedly to $d = x - y$, using coprimality of each next modulus with the preceding product.

N3.3 Every $x + \ell M$ satisfies the same congruences, and every solution has this form.

Export E3/v1. The solution set is $x + M\mathbb{Z}$, so the class modulo M is unique.

Acceptance gate. Both containments of the solution set are established; the product step has a coprimality certificate.

Targeted repair. If product divisibility was asserted without coprimality, restore the lemma before accepting uniqueness.

IV. CONTINUOUS MATHEMATICAL PROOF

Proof. Set $M = \prod_{i=1}^{k} m_i$ and $M_i = M/m_i$. We first show $\gcd(M_i, m_i) = 1$ (N1). If their gcd exceeded 1, it would have a prime divisor p by NT-02. Euclid's finite-product lemma would make p divide another modulus, contradicting pairwise coprimality. For $k = 1$, $M_i = 1$ and the claim is immediate. The same reasoning proves that any product of other moduli is coprime to m_i.

Bézout supplies $t_i, s_i \in \mathbb{Z}$ with $t_i M_i + s_i m_i = 1$. Define (N2)

$$x = \sum_{i=1}^{k} r_i M_i t_i.$$

For fixed j, every term with $i \neq j$ is divisible by m_j, whereas $M_j t_j \equiv 1 \pmod{m_j}$. Hence $x \equiv r_j \pmod{m_j}$, proving existence for every residue tuple.

For uniqueness (N3), let x, y be solutions and put $d = x - y$. Each m_i divides d. If coprime positive integers a, b divide d, write $d = ak$ and choose u, v with $ua + vb = 1$. Then $k = ud + vbk$, so $b \mid k$ and consequently $ab \mid d$. Repeated application, using the partial-product coprimality proved above, shows $M \mid (x - y)$. Conversely, since each m_i divides M, every integer $x + \ell M$ satisfies all congruences. The solution set is exactly one residue class modulo M.

□

V. WORKED ORCHESTRATION

Solve $x \equiv 2 \pmod 3$, $x \equiv 3 \pmod 5$, $x \equiv 2 \pmod 7$. N1 returns the following certificates.

m_i	M_i	t_i	Bézout certificate
3	35	2	$2 \cdot 35 - 23 \cdot 3 = 1$
5	21	1	$21 - 4 \cdot 5 = 1$
7	15	1	$15 - 2 \cdot 7 = 1$

N2 constructs $x = 2 \cdot 35 \cdot 2 + 3 \cdot 21 + 2 \cdot 15 = 233$. Reducing by $M = 105$ gives 23. Direct checks give remainders $2, 3, 2$, and N3 identifies every solution as $23 + 105\ell$, $\ell \in \mathbb{Z}$.

A. Boundary and hypothesis audit

- For $k = 1$, the construction is simply $x = r_1$ modulo m_1. Negative residues are allowed.
- Moduli $6, 10, 15$ have collective gcd 1 but are not pairwise coprime. This does not meet the contract.
- The conditions $x \equiv 0 \pmod 2$, $x \equiv 1 \pmod 4$ are incompatible. The compatible pair $0 \pmod 2$, $2 \pmod 4$ has solutions $2 + 4\mathbb{Z}$, not one class modulo 8.

VI. ILLUSTRATIVE FAILURE AND REPAIR

This is a deliberately authored teaching trace. A diagnostic asks for correction; it cannot prove a predecessor by importing its descendant's conclusion. A revised export must be checked before its consumers are reused.

Candidate. N1 attempts the product-modulus construction for $m_1 = 2, m_2 = 4$.

Detection. Validation finds $\gcd(4, 2) = 2$; no t can satisfy $4t \equiv 1 \pmod 2$.

Correction. Knowledge restores the pairwise-coprime contract. Strategy classifies the input as outside this theorem rather than inventing an inverse.

Recheck. N2 and N3 are withheld for that input. For valid moduli, Execution rechecks every local certificate and the uniqueness induction.

Publication. Contract preserves the stated CRT. Extending to noncoprime moduli would require compatibility conditions and a least-common-multiple modulus, a different theorem.

VII. REUSABLE RESULT AND PROVENANCE

Reusable exports retain each node's version 1 statement, assumptions, and proof. The final export is the theorem in Section 1, proved relative to its declared imports.

Source record. Adapted from the original project Markdown:
`chinese-remainder-theorem.md`

Typography, workflow conventions, and component cards follow the workspace Meta-Math manuscript. This authored exposition does not claim formal certification.